

Lab 3: Logistic Regression, Log-Loss, and Gradient Descent

Continue working in today's lab notebook. Use the model you already trained (w_{scaled} , b_{scaled}) unless a part says otherwise.

Part A: Predicting on New Students

- Your model has never seen these 5 new students. Using your already-trained model, predict each one's probability of passing and their class label: Study Hours = [1.2, 3.8, 5.1, 6.4, 9.2]
- Display the results in a small table (hours, probability, predicted class). Then identify which student the model is LEAST confident about (i.e., probability closest to 0.5), and briefly explain why that makes sense given their study hours.

Part B: Threshold Sensitivity

- Using your trained model on the original dataset, classify the data using three different thresholds: 0.3, 0.5, and 0.7 (instead of always using 0.5). For each threshold, report the Accuracy and Confusion Matrix. In 2–3 sentences, explain what changing the threshold does to the number of false positives vs. false negatives.